

Order-Flow Imbalance and Short-Horizon Returns: A Five-Pass Empirical Study

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Abstract. This note tests whether order-flow imbalance (OFI) — the net imbalance between aggressive buying and aggressive selling — predicts short-horizon returns, across five successive tests spanning two data qualities and two assets. An OHLCV-derived proxy on SPY (13 then 52 days) showed no relationship that survived out-of-sample validation. Real trade-level data on BTCUSDT (6 then 26 days, 24 million individual trades) initially suggested a small, stable effect, but a 4.3× increase in sample size collapsed the effect and revealed the earlier apparent stability to be a small-sample artifact. **No validated OFI signal survives across any test performed.** The methodological path to that conclusion — including a self-identified statistical error and its correction — is presented in full, since it is the more informative result.

1. Motivation and Definition

Market microstructure theory holds that trades initiated by an aggressor (the side that crosses the spread rather than waiting passively) carry information: an aggressor is revealing urgency or private information, and market makers should widen or shift quotes in response. If this holds even weakly, the net imbalance between aggressive buy volume and aggressive sell volume in a short window — order-flow imbalance — should have some power to predict the very next price move.

Formally, for a window with aggressor buy volume V_b and aggressor sell volume V_s : $OFI = (V_b - V_s) / (V_b + V_s)$, bounded in $[-1, +1]$. This is the definition used wherever the aggressor side is directly observed (Passes 3–5). Where it is not observed — standard OHLCV bars, as with most free equity data — a proxy is required (Passes 1–2, defined in Section 2.1).

2. Data and Methods

2.1 Passes 1–2: OHLCV proxy on equity data

SPY 1-minute bars (Twelve Data), first 5,000 bars (13 trading days), then 20,000 bars (52 trading days, chained pulls). Aggressor side is not observed in OHLCV bars, so it is proxied: signed volume = $\text{sign}(\text{open} - \text{close}) \times \text{volume}$, normalized by a trailing 30-bar average volume for cross-bar comparability. Tested against same-day forward returns at 1, 2, 5 and 10-minute horizons, with a chronological 70/30 walk-forward split (first 70% in-sample, last 30% out-of-sample, never mixed).

2.2 Passes 3–5: real trade-level data

BTCUSDT trade records from Binance's public historical archive (data.binance.vision), which records every individual trade with an `is_buyer_maker` flag — the actual tick-rule classification of which side was the aggressor, not an approximation. Trades were aggregated into 1-minute buckets (buy/sell volume summed by aggressor side) and tested first on 6 days (6.19 million trades), then 26 days (24.0 million trades) to examine statistical power directly.

The declared model, fixed before each data expansion: next-minute return = $a + b \cdot OFI + c \cdot (\text{previous-minute return})$, with the previous-minute-return term included as a control for ordinary short-horizon momentum. Standard errors use the Newey-West (HAC) estimator, robust to the serial correlation directly confirmed present in 1-minute returns (see Section 3.2). Same walk-forward split as Passes 1–2.

3. Results

3.1 Equity proxy (Passes 1–2)

No horizon produced a relationship that replicated cleanly in-sample to out-of-sample. At 1 and 2 minutes, the correlation was small but nominally significant in-sample and then **reversed sign** out-of-sample — the standard signature of a spurious in-sample pattern. The 5-minute horizon was the closest to consistent (same sign both slices) but was not significant out-of-sample alone. All effect sizes were tiny ($|r| < 0.03$) even where nominally significant.

Horizon	In-sample r [95% CI]	Out-of-sample r [95% CI]	Consistent?
1 min	+0.017 [+0.000, +0.034]	-0.022 [-0.047, +0.004]	No - sign flips
2 min	+0.019 [+0.003, +0.036]	-0.015 [-0.040, +0.011]	No - sign flips
5 min	+0.025 [+0.009, +0.042]	+0.012 [-0.013, +0.038]	Same sign, OOS n.s.
10 min	+0.014 [-0.003, +0.030]	+0.027 [+0.001, +0.052]	In-sample n.s.

Table 1. SPY, 52 trading days, 20,000 1-minute bars, OHLCV-derived proxy signal.

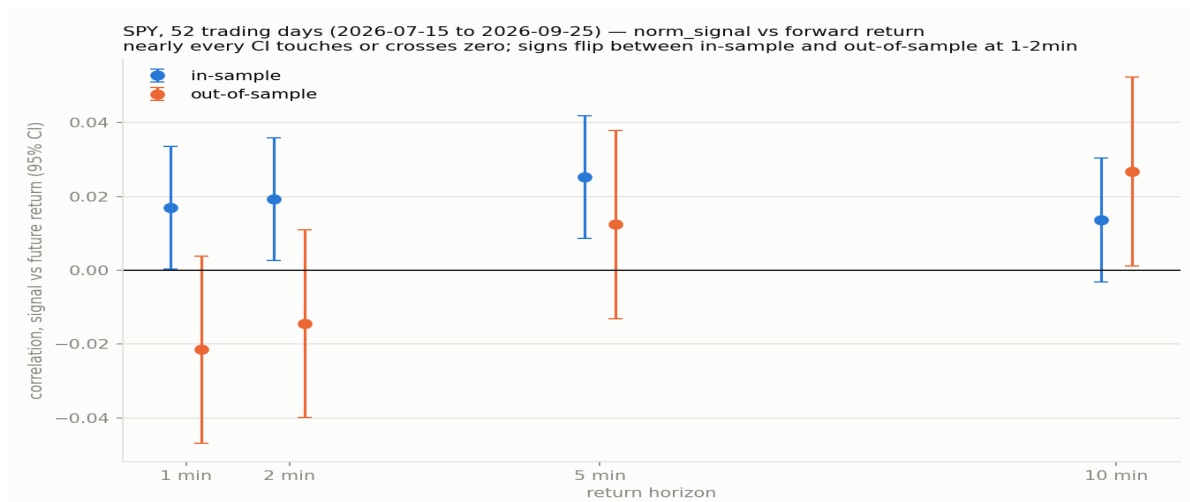


Figure 1. Correlation between the proxy signal and forward return, by horizon, in-sample vs. out-of-sample. Confidence intervals mostly touch or cross zero; where they do not, the sign is unstable across horizons.

3.2 Real trade-level data (Passes 3–5)

The first look at real data (6 days) was more encouraging: a naive test showed a significant out-of-sample correlation ($r = +0.061$), with the intuitive sign (more aggressive buying $\rightarrow$ price rises). Two rounds of audit were required before this could be trusted:

Audit 1 — momentum confound. The out-of-sample window had genuine positive return autocorrelation absent in-sample (ACF(1) of returns: -0.012 in-sample vs. $+0.086$ out-of-sample), so part of the apparent OFI effect could simply be that window's own price momentum. Controlling for this directly (adding the previous return as a regressor) made OFI's coefficient appear to vanish under ordinary least squares ($t = 0.93$).

Audit 2 — correcting the audit itself. Ordinary least squares standard errors assume independent residuals — an assumption already contradicted by the serial correlation just measured. Refitting the identical model with Newey-West (HAC) standard errors reversed the reading again: OFI's coefficient was small ($\beta \approx 0.00003$) but nearly identical across in-sample, out-of-sample and pooled data, and statistically significant in-sample ($t = 2.06$) and pooled ($t = 2.29$).

Audit 3 — the resolving test. A coefficient that stays the same size across samples while its significance test wavers is consistent with either a real, underpowered effect, or a small-sample coincidence. This was resolved directly with 4.3× more data (26 days, 24.0 million trades) under the identical, pre-specified model. The result: the coefficient did not sharpen — it collapsed (5–15× smaller) and lost cross-sample agreement.

Sample (6 days, 6.19M trades)	n (min)	OFI β	OFI t (HAC)
In-sample	6,047	+0.000032	2.06
Out-of-sample	2,591	+0.000024	0.96
Pooled	8,638	+0.000031	2.29

Sample (26 days, 24.0M trades)	n (min)	OFI β	OFI t (HAC)
In-sample	26,207	+0.0000017	0.23
Out-of-sample	11,231	+0.0000197	1.78
Pooled	37,438	+0.0000062	1.00

Table 2. BTCUSDT true aggressor-side OFI, momentum-controlled, HAC standard errors. Same specification, 6 vs. 26 days.



Figure 2. OFI's own coefficient (95% CI) at 6 vs. 26 days of data. The larger sample did not confirm the smaller sample's result — the estimate fell toward zero and the interval widened to include it.

4. Discussion

Across five tests, two data qualities, and two assets, no version of order-flow imbalance tested here shows a relationship with short-horizon returns that survives out-of-sample validation at adequate statistical power. The equity proxy failed outright, with sign reversals at every horizon where an in-sample pattern appeared. The real trade-level result is the more interesting failure: it survived one legitimate statistical correction (moving from OLS to HAC standard errors, justified by directly measured serial correlation) before failing the test that mattered — a 4.3× increase in sample size, run under the exact same pre-registered specification.

The value of this note is methodological rather than substantive: it demonstrates a discipline of treating an encouraging result as a hypothesis to be attacked, not a finding to be reported, and of holding a correction to the same standard as the original claim it corrected. A signal that requires this much scrutiny to rule out is, by construction, not a signal an execution strategy could safely be built on.

Limitations. All trade-level tests use a single exchange and a single pair (BTCUSDT); the proxy tests use a single equity ETF (SPY). Only a linear specification with one lag of momentum was tested. The trade tape captures executed trades, not resting order-book depth, so information in unfilled liquidity is not captured. Twenty-six days, while an improvement, is still short relative to the multi-year windows typical of published microstructure studies.

5. Next Steps

Candidates for extending this line of work, not pursued here: (i) incorporate order-book depth rather than only executed trades, to capture information in liquidity that was never filled; (ii) test across multiple pairs and a traditional-equity trade-level source (e.g. via a paid data vendor) to check whether the null result is asset-specific; (iii) test nonlinear or regime-conditional specifications, since the confound encountered here was itself regime-dependent (momentum present in one window, absent in another). This study's own infrastructure — the bucketing pipeline and the HAC-based testing harness — transfers directly to any of these.

Data sources: Binance historical trade archive (data.binance.vision); Twelve Data API. Full pass-by-pass detail, code and processed data available on request.